Pharmacological blockade of infection chronification modulates oxy-inflammation and prevents the activation of stress-induced premature senescence markers in schistosomiasis.

PMID: 39732412 | Indexed in PubMed

Chronic inflammation, oxidative stress, and DNA damage are observed in schistosomiasis and premature aging. However, the potential of these events to trigger stress-induced premature senescence (SIPS) throughout schistosomiasis progression remains overlooked, especially in response to the first-line pharmacological treatment. Thus, we investigated the relationship between oxidative stress and SIPS sentinel markers in untreated Schistosoma mansoni-infected mice and those receiving praziquantel (Pz)-based reference treatment. Swiss mice were randomized into 5 groups: uninfected (followed by 60- and 180-days post-infection), acutely (60 days) and chronically (180 days) infected untreated, and infected treated with Pz followed until 180 days. Our results indicated that infection chronification was accompanied by the worsening of hepatic granulomatous inflammation, increased number of granulomas, IL-4, TGF-β, reactive oxygen species (ROS) levels, fibrosis, hepatocytes DNA damage, upregulation in SA-β-gal activity, p16 and p21 gene expression, and hepatocytes proliferation down-regulation in the absence of telomeric shortening. These abnormalities were blocked by Pz treatment, which prevented infection chronification and the decline in hepatocytes proliferative potential, stimulating granulomatous inflammation resolution. Taken together, our findings provide the evidence that progressive fibrosis, sustained production of high ROS levels, marked DNA damage and decline in p16 and p21 expression are associated with hepatocytes replication attenuation in the chronic phase of S. mansoni infection. Thus, pharmacological blockade of infection and granulomatous inflammation is essential to prevent these premature senescence markers associated with hepatocytes replicative disorders, stimulating liver regeneration in schistosomiasis mansoni.

Treatment of urinary schistosomiasis: methodological issues and research needs identified through a Cochrane systematic review.

PMID: 19493363 | Indexed in PubMed

Guidelines recommend praziquantel (PZQ) for the treatment and control of schistosomiasis, with no real alternative. Metrifonate was still widely used against Schistosoma haematobium in the 1990s, and then withdrawn. Experimental studies and clinical trials suggest that artemisinin compounds are active against S. haematobium. In a Cochrane systematic review assessing the efficacy and safety of drugs for treating urinary schistosomiasis, 24 randomized controlled trials (n=6315 individuals) met our inclusion criteria. These trials compared a variety of single agent and combination regimens with PZQ, metrifonate or artemisinin derivatives. The review confirmed that both the standard recommended doses of PZQ (single 40 mg/kg oral dose) and metrifonate (3x7.5-10 mg/kg oral doses administered fortnightly) are efficacious and safe in treating urinary schistosomiasis, but there is no study comparing these two regimens head-to-head. There is currently not enough evidence to evaluate artemisinin compounds. Most of the studies included in the Cochrane systematic review were insufficiently powered, lacked standardization in assessing and reporting outcomes, and had a number of methodological limitations. In this paper we discuss the implications of these findings with respect to public health and research methodology and propose priority research needs.

Clinical efficacy and tolerability of praziquantel for intestinal and urinary schistosomiasis-a meta-analysis of comparative and non-comparative clinical trials.

PMID: 25412105 | Indexed in PubMed

Extensive use of praziquantel for treatment and control of schistosomiasis requires a comprehensive understanding of efficacy and safety of various doses for different Schistosoma species. A systematic review and meta-analysis of comparative and non-comparative trials of praziquantel at any dose for any Schistosoma species assessed within two months post-treatment. Of 273 studies identified, 55 were eligible (19,499 subjects treated with praziquantel, control treatment or placebo). Most studied were in school-aged children (64%), S. mansoni (58%), and the 40 mg/kg dose (56%); 68% of subjects were in Africa. Efficacy was assessed as cure rate (CR, n=17,017) and egg reduction rate (ERR, n=13,007); safety as adverse events (AE) incidence. The WHO-recommended dose of praziquantel 40 mg/kg achieved CRs of 94.7% (95%CI 92.2-98.0) for S. japonicum, 77.1% (68.4-85.1) for S. haematobium, 76.7% (95%CI 71.9-81.2) for S. mansoni, and 63.5% (95%CI 48.2-77.0) for mixed S. haematobium/S. mansoni infections. Using a random-effect meta-analysis regression model, a dose-effect for CR was found up to 40 mg/kg for S. mansoni and 30 mg/kg for S. haematobium. The mean ERR was 95% for S. japonicum, 94.1% for S. haematobium, and 86.3% for S. mansoni. No significant relationship between dose and ERR was detected. Tolerability was assessed in 40 studies (12,435 subjects). On average, 56.9% (95%CI 47.4-67.9) of the subjects receiving praziquantel 40 mg/kg experienced an AE. The incidence of AEs ranged from 2.3% for urticaria to 31.1% for abdominal pain. The large number of subjects allows generalizable conclusions despite the inherent limitations of aggregated-data meta-analyses. The choice of praziquantel dose of 40 mg/kg is justified as a reasonable compromise for all species and ages, although in a proportion of sites efficacy may be lower than expected and age effects could not be fully explored.

The role of artesunate for the treatment of urinary schistosomiasis in schoolchildren: a systematic review and meta-analysis.

PMID: 23265611 | Indexed in PubMed

Efficacy of artemisinin derivatives alone or in combination compared to praziquantel alone for the treatment of urinary schistosomiasis in schoolchildren. Randomized clinical trials comparing praziquantel with artemisinin derivatives in the treatment of urinary schistosomiasis in schoolchildren were included. Medline, EMBASE, LILACS, CENTRAL, African Index Medicus, and Scielo were searched. We also analyzed the abstracts of the main conferences on infectious diseases and tropical medicine during the years 2009-2011. Google Scholar and OpenSIGLE were also searched. The last search was performed in July 2012. The primary endpoint was the cure rate. The main outcome data were retrieved using a standardized form; three independent researchers (WP, HC, and SS) performed the search, retrieved data, and evaluated the risk of bias. Disagreements were resolved by discussion. Risk ratios were used and heterogeneity was evaluated. A fixed or random-effects model was used according to the results of heterogeneity testing. An intention-to-treat analysis was done. Data were analyzed using Revman 5·0·24 (Copenhagen: The Nordic Cochrane Centre). Seven studies were selected for full text review and only five studies were finally included. The cure rate for praziquantel was superior to that of artesunate (RR: 1·66; 95% CI: 1·18-2·33). Artesunate was not clearly superior to placebo (artesunate versus placebo, RR: 3·21; 95% CI: 0·50-20·74). Combination of artesunate with praziquantel could prove more beneficial than praziquantel alone (RR: 1·15; 95% CI: 1·01-1·31). The frequency of adverse events was equivalent for

both drugs (praziquantel versus artesunate, RR: 1·11; 95% CI: 0·80-1·55). Our meta-analysis showed that praziquantel was significantly more effective than artesunate for the treatment of urinary schistosomiasis in schoolchildren. Artesunate at best had a marginal role in combination therapy.

A comparative study on the efficacy of praziquantel and albendazole in the treatment of urinary schistosomiasis in Adim, Cross River State, Nigeria.

PMID: 28911129 | Indexed in PubMed

Praziquantel (PZQ) is the current drug of choice for the treatment of urinary schistosomiasis in endemic areas. It is very efficacious, although the potential for the development of resistance has been reported in some endemic areas among human subjects and in animal studies. Its' limitation include high cost and administration of multiple numbers of tablets. Albendazole (ALB) is used in the treatment of intestinal helminths infection. It is a broad-spectrum single-dose antihelminthic with an excellent cure rate and safety criteria. Currently, it is not routinely used for the treatment of urinary schistosomiasis. Urine samples collected from 596 pupils aged between 2 and 16 years were processed and examined for the presence of ova of Schistosoma haematobium using a standard filtration technique. A total of 100 infected subjects were treated with a standard dose of PZQ (40 mg/kg body weight), while another group of 96 infected subjects were treated with ALB (400 mg for individuals above 3 years). A post-treatment study was conducted 1 month after treatment to assess their cure rate. The prevalence of S. haematobium infection in the study area was 32.8% (196/596). More males were infected (44.2%) (122/276) than females (23.1%) (74/320). The difference in the prevalence rate of infection by gender was statistically significant ($X2=15.7>3.841$, $p<0.05$). The highest prevalence of infection was observed among subjects aged 14-16 years (42.1%) (32/76), while those aged 5-7 years had the least prevalence (23.7%) (38/160). There was no statistically significant difference in the prevalence of urinary schistosomiasis by age of the subjects (X2=5.990.05). PZQ gave a higher cure rate of 78.0% (78/100) compared with ALB (68.7%) (66/96). There was no statistically significant difference in the cure rate obtained with both drugs ($X2=0.355>0.282$, $p>0.05$). The intensity of egg excretion was greatly reduced in subjects who were not cured by the two drugs. The findings of this study suggest the use of ALB for the treatment of urinary schistosomiasis. We recommend further assessment of the efficacy of the drug in an area with higher morbidity of urinary schistosomiasis than the present study area.

Evidence for in vitro and in vivo activity of the antimalarial pyronaridine against Schistosoma.

PMID: 34166393 | Indexed in PubMed

Schistosomiasis is highly prevalent in Africa. Praziquantel is effective against adult schistosomes but leaves prepatent stages unaffected-which is a limit to patient management and elimination. Given the large-scale use of praziquantel, development of drug resistance by Schistosoma is feared. Antimalarials are promising drugs for alternative treatment strategies of Schistosoma infections. Development of drugs with activity against both malaria and schistosomiasis is particularly appealing as schistosome infections often occur concomitantly with malaria parasites in sub-Saharan Africa. Therefore, antiplasmodial compounds were progressively tested against Schistosoma in vitro, in mice, and in a clinical study. Amongst 16 drugs and 1 control tested, pyronaridine, methylene blue and 5 other antimalarials were

highly active in vitro against larval stage schistosomula with a 50% inhibitory concentration below 10 µM. Both drugs were lethal to ex vivo adult worms tested at 30 µM with methylene blue also active at 10 µM. Pyronaridine treatment of mice infected with S. mansoni at the prepatent stage reduced worm burden by 82% and cured 7 out of 12 animals, however in mice adult stages remained viable. In contrast, methylene blue inhibited adult worms by 60% but cure was not achieved. In an observational pilot trial in Gabon in children, the antimalarial drug combination pyronaridine-artesunate (Pyramax) reduced S. haematobium egg excretion from 10/10 ml urine to 0/10 ml urine, and 3 out of 4 children were cured. Pyronaridine and methylene blue warrant further investigation as candidates for schistosomiasis treatment. Both compounds are approved for human use and evidence for their potential as antischistosomal compounds can be obtained directly from clinical testing. Particularly, pyronaridine-artesunate, already available as an antimalarial drug, calls for further clinical evaluation. ClinicalTrials.gov Identifier NCT03201770.

Efficacy and Safety of Moxidectin, Synriam, Synriam-Praziquantel versus Praziquantel against Schistosoma haematobium and S. mansoni Infections: A Randomized, Exploratory Phase 2 Trial.

PMID: 27636542 | Indexed in PubMed

Schistosomiasis affects millions of people, yet treatment options are limited. The antimalarial Synriam (piperaquine 150 mg/arterolane 750 mg) and the anthelminthic moxidectin revealed promising antischistosomal properties in preclinical or clinical studies. We conducted two single-blind, randomized exploratory Phase 2 trials in Schistosoma mansoni and S. haematobium-infected adolescents in northern and central Côte d'Ivoire. Our primary endpoints were cure rates (CRs) and egg reduction rates (ERRs) based on geometric mean and safety. Each subject was asked to provide two stool samples (S. mansoni trial) for Kato-Katz analysis or three urine samples (S. haematobium trial) for urine filtration and one finger prick for malaria screening at baseline and follow-up. Participants were randomly assigned to either moxidectin, Synriam, Synriam plus praziquantel or praziquantel. 128 adolescents (age: 12-17 years) were included in each study. Against S. haematobium moxidectin and Synriam revealed low efficacy. On the other hand, Synriam plus praziquantel and praziquantel yielded CRs of 60.0% and 38.5% and ERRs of 96.0% and 93.5%, respectively. CRs observed in the treatment of S. mansoni were 13.0%, 6.7%, 27.0%, and 27.6% for moxidectin, Synriam, Synriam plus praziquantel and praziquantel, respectively. ERRs ranged from 64.9% (Synriam) to 87.5% (praziquantel). Synriam and moxidectin show low efficacy against S. haematobium, hence an ancillary benefit is not expected when these drugs are used for treating onchocerciasis and malaria in co-endemic settings. Further studies are needed to corroborate our findings that moxidectin and Synriam show moderate ERRs against S. mansoni.

Drugs for treating urinary schistosomiasis.

PMID: 18646057 | Indexed in PubMed

Urinary schistosomiasis causes long-term ill-health. This review examines the various treatment options and newer drugs. To evaluate antischistosomal drugs, used alone or in combination, for treating urinary schistosomiasis. In August 2007, we searched the Cochrane Infectious Diseases Group Specialized Register, CENTRAL (The Cochrane Library 2007, Issue 3), MEDLINE, EMBASE, LILACS, mRCT, and reference lists of articles. We also contacted experts in schistosomiasis research. Randomized and quasi-randomized controlled trials of praziquantel, metrifonate, artemisinin derivatives, or albendazole, alone or in combination, versus placebo, different doses, or other antischistosomal drugs for treating urinary schistosomiasis. One author extracted data, and assessed eligibility and methodological quality, which were cross-checked by a second person. Dichotomous outcomes were combined using risk ratio (RR), and continuous data were combined using weighted mean difference (WMD); both presented with 95% confidence intervals (CI). Twenty-four trials (6315 participants) met the inclusion criteria. Compared with placebo, participants receiving metrifonate had fewer parasitological failures at follow up at one to three months (1 trial) and three to 12 months (3 trials). Egg reduction rate was over 90%, and no adverse events were reported (1 trial). One metrifonate dose was inferior to three doses given fortnightly (both used 10 mg/kg). Praziquantel (standard single 40 mg/kg oral dose) was more effective than placebo at reducing parasitological failure at one to three months' follow up and three to 12 months. Egg reduction rates were improved with praziquantel (over 95% versus 5.3% to 64% with placebo). Mild to moderate adverse events were recorded in two trials. A comparison of metrifonate (10 mg/kg x 3, once every 4 months for one year) with praziquantel (standard dose) showed little difference in parasitological failure. For praziquantel, there was no significant difference in effect between 20 mg/kg x 2, 30 mg/kg x 1, and 20 mg/kg x 1, and the standard dose for all outcomes. One small trial of artesunate showed no obvious benefit compared with placebo, and the artesunate-praziquantel combination was similar to praziquantel alone. Praziquantel and metrifonate are effective treatments for urinary schistosomiasis and have few adverse events. Metrifonate requires multiple administrations and is therefore operationally less convenient in community-based control programmes. Evidence on the artemisinin derivatives is currently inconclusive, and further research is warranted on combination therapies. We suggest metrifonate be reconsidered for the WHO Model List of Essential Medicines.

Efficacy of artesunate in the treatment of urinary schistosomiasis, in an endemic community in Nigeria.

PMID: 15257799 | Indexed in PubMed

The efficacy and tolerability of oral artesunate for the treatment of urinary schistosomiasis was assessed among schoolchildren aged 5-18 years in Adim community, Nigeria. Overall, 500 children, randomly selected from those attending the Presbyterian primary school, were each invited to provide two consecutive urine samples. Using standard parasitological procedures, Schistosoma haematobium ova were found in the samples from 145 (29.0%) of the subjects. Most (87) of the infected subjects were then treated orally with artesunate, using two doses, each of 6 mg/kg, given 2 weeks apart. When the treated children were re-examined 4 weeks after the second dose of artesunate, 61 (70.1%) appeared egg-negative and were therefore considered cured. Post-treatment, the geometric mean egg count (GMEC) for the treated subjects who were not cured was significantly lower than the pre-treatment GMEC for all the treated subjects, with log10[(eggs/10 ml urine) + 1] values of 0.9 v. 1.75 (t = 4.45; P or = 10 years was slightly higher than that among the younger subjects. It was lowest for the heavier subjects (70% for those weighing 41-50 kg) and highest (79%) for the subjects who weighed 31-40 kg. The artesunate was well tolerated. This observation of a therapeutic effect of artesunate against S. haematobium in Nigeria confirms recent observations from Senegal. In the Adim community at least, it would be more cost-effective to treat urinary schistosomiasis with artesunate than with praziquantel. The wide-spread use of artesunate against schistosomiasis has to be considered carefully, however, if it is not to compromise the efficacy of the drug as an antimalarial, by increasing the risk of resistance developing in local Plasmodium.

Efficacy of a combination of praziquantel and artesunate in the treatment of urinary schistosomiasis in Nigeria.

PMID: 18838149 | Indexed in PubMed

The combined effects of praziquantel and artesunate in the treatment of urinary schistosomiasis were assessed among 312 randomly selected schoolchildren aged 4-20 years in Adim community, Nigeria. In the preliminary screening, infection was confirmed in 327 (38.5%) of the 850 subjects screened. Infected subjects who reported for treatment were then divided into six treatment groups of 52 subjects each; 44 subjects in each group completed their treatment regimens and submitted their urine for post-treatment assessment. Praziquantel and artesunate were administered orally at 40 mg/kg and 4 mg/kg body weight, respectively. Adverse effects due to drug reactions were assessed 72 h after medication and all perceived episodes of illness were treated. Morbidity indicators were assessed 56 days after the final dose of the drug regimens. All treatment regimens were well tolerated. The cure rates were 72.7% in the praziquantel plus placebo-treated group and 70.5% in the artesunate plus placebo group, while the artesunate plus praziquantel group had the highest cure rate (88.6%). Haematuria and proteinuria were extensively reduced after treatment with the three drug regimens. This study confirmed that the treatment of urinary schistosomiasis with the combination of praziquantel and artesunate is safe and more effective than treatment with either drug alone.